Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “medium” and “wave” to compare your model or result. Sentence starters are optional.

medium: the material through which a mechanical wave travels

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

01 Bring back an earlier idea

A converging lens has f=15 cm and d_o=6 cm.
Use $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$.
Find d_i, magnification and image kind.

Word help: lens = a transparent element that changes light-ray directions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $d = v\Delta t$; $\Delta t = d/v$.
Use the supplied wave speed for its medium and wave type.
These classroom models are not real hazard forecasts.

Worked reasoning: In a simple model, a wave amplitude falls from 4 to 2 units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

Your turn: Draw the two waves and two corresponding relative-intensity bars.

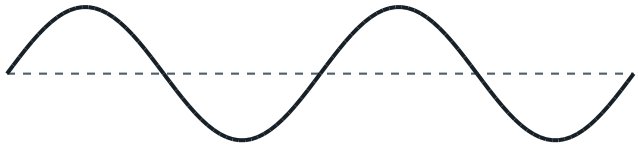
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw a disturbance traveling through water and mark local water motion. Why does wave arrival not mean the original water crossed the whole ocean?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Solve and justify

Supplied seismic-speeds: P=6 km/s, S=3.5 km/s. At a station 210 km away, find arrival times and S-P delay.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why does knowing wave arrival time alone not establish local damage?
Name other evidence.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Light moves from a transparent medium with n=1.7 into air (n=1).
Find critical angle using $\sin\theta_c = 1/n$.
Describe incidence above it in the ideal model.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A shallow-water model uses $v=\sqrt{(gh)}$. With g=10 and h=100 m, find speed and time for 10 km.
State a limitation.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A signal has intensity 8 W/m² at 10 m from an ideal point source.
Find intensity at 30 m and total radiated power.
Explain one limitation for ground waves.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3